

— COMPLETE LEARNING ROADMAP —

# Full Stack Data Science

Advanced Generative AI

Advanced Agentic AI

Full Stack with  
Deployment

— a complete end-to-end curriculum.

11

MODULES

5

PHASES

70+

TOPICS

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POTENTIAL

## PHASE 01 FOUNDATIONS



01

### Python

The universal language of data and AI. Master syntax, data structures, OOP, and the scientific Python ecosystem.

pandas numpy matplotlib OOP decorators



02

### EDA & Statistics

Transform raw data into insights. Exploratory analysis, probability theory, and statistical inference for data scientists.

hypothesis testing distributions correlation Bayesian

## PHASE 02 MACHINE LEARNING CORE



03

### Machine Learning

Core algorithms and model building — from regression to ensemble methods. The backbone of every AI system.

scikit-learn XGBoost feature eng. cross-val



04

### ML-Ops & CI/CD

Move models from notebooks to production. Versioning, pipelines, monitoring, and continuous delivery for ML systems.

MLflow DVC Docker GitHub Actions FastAPI

PHASE 03

DEEP AI & NEURAL SYSTEMS

05

### AI — NLP · DL · NN

Neural architectures, deep learning pipelines, and natural language processing from scratch to Transformers.

PyTorchTransformersBERTCNNs

06

### AI-Ops

Operationalize AI models at scale. Infrastructure, monitoring, model governance, and reliability engineering for AI.

KubernetesPrometheusGrafanaSeldon

07

### Advanced Generative AI

GANs, VAEs, diffusion models, and multimodal generation. Build systems that create images, audio, code, and more.

DiffusionGANsVAEStable Diffusion

PHASE 04

LLMS · PROMPT ENGINEERING · ADVANCED AGENTIC AI

08

### LLM & LLM-Ops

Fine-tune, evaluate, and deploy large language models. RLHF, PEFT methods, quantization, and production LLM stacks.

LoRARRLHFvLLMLangChainRAG

●●● Expert

TOPICS ↓

09

### Prompt Engineering

The art and science of communicating with LLMs. From zero-shot to chain-of-thought, adversarial prompts, and systematic evaluation.

CoTfew-shotReActDSPyred-teaming

●● Intermediate

TOPICS ↓

10

### Advanced Agentic AI

Autonomous multi-agent systems, tool use, and the Model Context Protocol. The frontier of AI engineering.

MCPAutoGenCrewAItool use

AI Tools

●●● Expert

TOPICS ↓

PHASE 05

FULL STACK WITH DEPLOYMENT

11

### Full Stack with Deployment

Build and ship complete AI-powered web applications — from backend APIs to front-end UIs to cloud deployment. Tie every skill together into production-ready full stack systems.

FastAPIReactStreamlitDockerAWS / GCPCI/CDPostgreSQLRedis

●●● Expert

TOPICS ↓

KODI PRAKASH SENAPATI

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